

JAKUB T SZWEDO

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EDUCATION

Bachelor of Science in Computer Science

May 2028

University of Illinois Chicago | Chicago, IL

- GPA: 3.72 | Cum laude, National Honor Society
- Relevant coursework: CS 361-Systems Programming, CS 251-Data Structures, CS 261-Machine Organization
- Minecraft Server Admin for LUG (Linux User Group)

EXPERIENCE

Software Developer Intern, Web & Embedded Systems

June 2026 – August 2026

Sterling Solutions (Discover Partner Institute) | mHUB, Chicago, IL

- Strengthened file sharing security by implementing **password authentication**, **Apache** directory restrictions (**.htaccess**), and **password hashing** via **PHP**, fixing the problem that allowed files to be accessed by anyone.
- Reviewed an existing **PHP** codebase for security vulnerabilities, and integrated file share notifications using **PHPMailer** over **SMTP/TLS**.
- Diagnosed a connection bug in a device's **TCP** networking layer, tracing it to a **thread safety** issue and fixing it by preventing **concurrent writes** and **rate limiting**.
- Built **automatic reconnect** and **retry logic** to recover dropped **TCP** connections between server and devices.

Teaching Assistant for Languages and Automata

January 2026 – May 2026

University of Illinois Chicago | Chicago, IL

- Ran weekly office hours to help students on **finite automata**, **regular expressions**, and **context-free grammars**.
- Led weekly lab sessions for 25+ students, providing hands on practice with **JFLAP** and content covered in lecture.

IT Intern - Infrastructure and Systems Administration

July 2025 – August 2025

University of Illinois Chicago | Chicago, IL

- Streamlined workstation rollouts by building **VMs** on **Proxmox VE** and automating imaging via **FOG Server**, leading a department wide migration with zero downtime for staff.
- Resolved hardware and software problems through **TDX ticketing**, troubleshooting system failures, and maintaining up-to-date network documentation.
- Strengthened **AV security** by configuring 10+ **Crestron XiO Cloud** panels with **SSL/TLS certificates** and **Google Calendar** integration.
- Managed the migration of 50+ **Cisco IP** phones to **Zoom Phone**, tracking device inventory (MAC/IP) and troubleshooting connectivity issues.

PROJECTS

Self-Hosted VPN Infrastructure with Cloud Relay Architecture

July 2026 – Present

- Migrated a personal VPN network from a third-party service to a self-hosted **Headscale** server.
- Deployed a \$0/month **Google Cloud** relay (**rathole** over TCP) so the self-hosted VPN server could be reached from the public internet without exposing the home network's IP address or opening any router ports.
- Fixed a connection problem where **Cloudflare** dropped a needed **HTTP** header, then redesigned the relay around it.
- Secured the cloud server by limiting **SSH** access, capping outgoing network traffic at 900MB/month, and turning on automatic security updates.
- Found and fixed a **DNS** setting that had been breaking network access across several containers.

Self-Hosted Media Automation Homelab

March 2025 – Present

- Built and ran a home server using **Proxmox VE**, managing 10+ **Linux containers** and services.
- Set up an automated media system using **Jellyfin**, with automated content indexing, metadata management, and request handling integrations across multiple containerized services.
- Built an **nftables** kill switch to force one device to only connect through a **VPN**, and fixed a hidden **IPv6** leak.
- Set up network wide **DNS** filtering and monitoring with **Pi-hole**.
- Patched two known **CVEs** on a public facing **Jellyfin** server, then deployed **Wazuh** SIEM and checked its findings against **Debian's** security tracker, removing two more unused packages tied to roughly 30 flagged **CVEs**.

SKILLS

- **Programming & Scripting:** C/C++, JavaScript, HTML, CSS, Bash, PHP
- **Systems, Networking & Cloud:** Proxmox VE, Linux, VMs, Apache, nftables, TCP/DNS/SSH, TLS/SSL, VPN (Headscale), IPv6, Google Cloud, Cloudflare, Jellyfin, Pi-hole, SMTP/HTTP, CVE Remediation
- **Certifications & Languages:** Cisco Networking Academy (Network Security, Data Privacy), TDX Ticketing, Apple GSX, AutoCAD | Fluent in English & Polish